??
$\rho(\mathbf{r})$
$E[\rho] = T_s[\rho] + \int v_{\text{ext}}(\mathbf{r})\rho(\mathbf{r})\, d\mathbf{r} + J[\rho] + E_{\text{xc}}[\rho] + E_{\text{NN}}.$
(1)
T_s
v_{ext}
J
E_{xc}
E_{NN}
E_{xc}
??
??
??
d
ρ
??
+
$\Delta E_i = E_i - E_{\text{ref}}.$
(2)
$f(\mathbf{r}) = \left(\frac{\partial \rho(\mathbf{r})}{\partial N}\right)_{v_{\text{ext}}}.$
(3)
??
+(r) =
$\rho_{N+1}(\mathbf{r}) -$
$\rho_N(\mathbf{r}),$
$f^-(\mathbf{r}) =$
$\rho_N(\mathbf{r}) -$
$\rho_{N-1}(\mathbf{r}),$
$f^0(\mathbf{r}) =$
$12\left[\rho_{N+1}(\mathbf{r}) - \rho_{N-1}(\mathbf{r})\right].$
f^+
f^-
f^0
I
A
$I \approx$
$-\varepsilon_{\text{HOMO}}$
$A \approx$
$-\varepsilon_{\text{LUMO}}$
??
μ
χ
$\mu \approx -\frac{I+A}{2}, \chi = -\mu.$
(4)
$\eta \approx$
$I -$
A
$S = \frac{1}{\eta}, \omega = \frac{\mu^2}{2\eta}.$
(5)
??
$\eta_{1/2} =$
$(I -$
$A)/2$
$S \approx$
$1/(2\eta_{1/2})$
$\omega \approx$
$\mu^2/(4\eta_{1/2})$
I
A
μ
χ
η
ω
S
-1
$V(\mathbf{r}) = \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} - \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}'.$
(6)